Cogs 209 Mini-Project Proposal

Lead Author: Bryan Zhu

Co-authors: Travis Ramirez, Jiaguan Pan, Dylan Walmsley

Title: **Speed-Dating Behaviors: An Investigation of Demographics vs. Preferences**

Research question.

The choice of a romantic partner remains a relevant question since the dawn of humanity, and understanding what factors guide a person to choosing their significant other remains an open-ended question. Furthermore, understanding how consistent one's expected preferences versus the attributes actually found in their partner can be important in understanding how one's background influences their preferences versus what influences their final decisions. Ultimately, we aim to investigate whether one's demographic background serves as a greater predictor of preferences versus their expected preferences and how those may be closely related. We propose to use speed dating data from experiments run from 2002-2004 and test various inference methods to first understand dating habits based on demographics. We hypothesize that dimensionality reductions such as PCA or even Lasso-regularized predictive models can identify key features that determine whether two partners will "match" with each other, which provides guidance in feature-selection and engineering for classification models. Furthermore, we hypothesize that non-parametric classifiers like K-Nearest Neighbors will prove to be limited when moving from broader demographic features to higher-level interactions, necessitating potential deeper models such as Neural Networks.

Data/materials.

This dataset was found under the name SpeedDating on OpenML.org. This data was gathered from participants in experimental speed dating events from 2002-2004. During the events, the attendees would have a four-minute "first date" with every other participant of the opposite sex. At the end of their four minutes, participants were asked if they would like to see their date again. The dataset also includes questionnaire data gathered from participants at different points in the process. The data is available in this Google Drive folder. The folder contains the following file:

File name	Format	Description
speed-dating.csv	Csv file	Data table with 8,378 rows (one speed-date interaction) and 123 columns (variables describing that interaction) Each row corresponds to a single speed-date interaction: one focal participant's record for one partner in one event wave. The first column, has_null, indicates whether key fields on that row are

		missing; the remaining columns hold demographics and pairing context (e.g. ages, race, field), stated preferences and trait weights, ratings (how the focal person scores the partner, self-ratings, and partner's scores of them), interest/hobby levels and overlap, expectations about the event, and outcomes for that date (liking, guessed interest, whether they had met before, each person's decision, and mutual match).

When using this dataset, treat each row as one directional mini-date (one person's record for one partner), not an independent observation: the same participants and waves appear many times, so standard errors and tests that assume independent rows are usually too optimistic unless you cluster (e.g. by person or wave), use mixed models, or aggregate to the level that matches the research question. Pairs may also appear twice (A→B and B→A), which matters for how you define the unit of analysis. Many fields exist in raw and binned (d_*) forms—avoid using both redundantly. Missing values and the has_null flag should be handled explicitly. Finally, be careful about which variables are available for prediction (e.g. post-date ratings vs pre-date attributes), and expect sparse categories if you slice demographics finely.

Course impact/relevance.

This project explores inferential techniques aiming to understand the influences between demographic/preferential features versus ultimate outcomes for matching. In particular, the project explores dimensionality reduction and feature exploration, and the conclusions drawn from these can be used for refining classification methods via supervised learning.

Outcome(s).

The outcome of this project will be a collection of features that result from dimensionality reduction techniques, including Lasso regression and PCA to capture statistical variance, which could be evaluated using cross-validation. These features can also be used to compare and contrast classification models like Bayesian Classifiers and K-nearest neighbors to see if dimensionality-reduced features improve classification performance.